

Notation and Legend

- **PK (Primary Key):** Unique record identifier for each entity.
- **FK (Foreign Key):** Reference to a primary key in a parent table to maintain referential integrity.
- **1 : M (One-to-Many):** One record in a table corresponds to multiple records in another table.
- **1 : 1 (One-to-One):** Each record in a table corresponds to exactly one record in another table.

2. Entities Description

- **Airport:** Stores general airport details including name, geographic location (country, state, city), and timestamps.
- **Airline:** Stores airline profiles, unique company codes, names, and registration countries.
- **Flight:** Tracks scheduled and actual flights, connecting departure/arrival airports with operating airlines.
- **Passenger:** Maintains traveler identity records, citizenship, residence, and passport details.
- **Booking:** Represents flight reservations made by passengers, containing status and ticket prices.
- **BookingFlightChange:** Audit table that records flight booking modification history and reasons.
- **BoardingPass:** Stores issued seat numbers, boarding times, and references the confirmed booking.
- **Baggage:** Records registered luggage units and their weight tied to a flight booking.
- **BaggageCheck:** Stores baggage screening results at airport security.
- **SecurityCheck:** Stores airport passenger screening and security checkpoint logs.

3. Relationships and Cardinality

- **Airline $\rightarrow$ Flight (1:M):** An airline can operate multiple flights; each flight belongs to one airline.
- **Airport $\rightarrow$ Flight (1:M):** Airports link to flights as departure and arrival points.
- **Flight $\rightarrow$ Booking (1:M):** A flight accommodates multiple passenger bookings.
- **Passenger $\rightarrow$ Booking (1:M):** A passenger can make multiple bookings over time.
- **Booking $\rightarrow$ BoardingPass (1:1):** Each booking is issued exactly one boarding pass.
- **Booking $\rightarrow$ Baggage (1:M):** A booking allows registering multiple pieces of baggage.
- **Baggage $\rightarrow$ BaggageCheck (1:1 / 1:M):** Each baggage item undergoes terminal screening.
- **Passenger $\rightarrow$ SecurityCheck (1:M):** A passenger completes security verification before departure.
- **Booking $\rightarrow$ BookingFlightChange (1:M):** Tracks multiple status modifications for a booking.

